

SUBJECT

NACA 4412 finite wing

source geometry: naca4412_wing.stl

OBJECTIVE	Solve the external aerodynamics of the supplied NACA 4412 finite-wing geometry at cruise Reynolds number. Select the appropriate turbulence model and solver, gate the mesh on quality, and report the lift and drag coefficients with confidence envelopes.
SOLVER & MODEL	OpenFOAM, k-omega SST steady RANS
ISSUED (UTC)	2026-07-23T20:52:17Z

RESULT

VALIDATED

Drag coefficient

0.01892 ± ±1.9e-05 over the final 24 iterations

95% confidence interval

Basis: within 37% of Abbott & von Doenhoff, Theory of Wing Sections (1959), NACA 4412 section data (section Cd ~ 0.006-0.009), Cd 0.03 (band ±40%)

Lift coefficient	0.2689 ±0.0002
Drag vs experiment	Cd 0.0189 vs 0.03 37% apart, ±40% band
Mesh	337,334 cells max non-orthogonality 49.9°, max skewness 3.32

UNCERTAINTY - ASME V&V 20 THREE-CHANNEL

CHANNEL	VALUE	STATE
input	2e-05	quantified
the ± band is the statistical spread of the solved force over the averaging window, settled-state scatter; freestream speed and fluid properties are taken as specified exactly		
numerical	—	not quantified
study pending: no matching refinement study for this setup		
model	—	not quantified
kOmegaSST closure error not estimated for this body		

PROVENANCE - EVIDENCE-BUNDLE SEAL (SHA-256)

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23d7e7e8074ac074e681bd9fad722a80

Mission geometry-study-naca4412_wing - Certificate C-2026-7069